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Effects of curcumin on serum inflammatory biomarkers in patients with knee osteoarthritis: a systematic review and meta-analysis of randomized controlled trials

Han-Chien Hsueh¹, Guan-Ru Ho², Shiang I. Tzeng³, Kai-Hsiang Liang⁴ and Yi-Shiung Horng^{2,5*}

Abstract

Background Curcumin is a potent anti-inflammatory and antioxidant herbal medicine that has been shown to exert beneficial effects on knee osteoarthritis (OA). Previous studies revealed that curcumin had a superior effect compared to control treatments in terms of pain intensity and other patient-reported outcomes, including the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), Knee Injury and Osteoarthritis Outcome Score (KOOS), visual analog scale (VAS), and Japanese Knee Osteoarthritis Measure (JKOM). However, it remains unclear whether curcumin has beneficial effects on serum inflammatory biomarkers. This systematic review and meta-analysis (SRMA) aimed to evaluate the effects of curcumin on changes in serum inflammatory biomarkers in patients with knee OA.

Methods The PubMed, MEDLINE, Embase, Web of Science, Cochrane Library, and EBSCOhost databases were systematically searched from inception to March 28, 2025 to identify all randomized clinical trials (RCTs) that examined the use of curcumin for treating knee OA. Data related to serum inflammatory biomarkers, including C-reactive protein (CRP) levels, the erythrocyte sedimentation rate (ESR), tumor necrosis factor (TNF-alpha) levels, interleukin (IL)-1beta levels, IL-6 levels, and prostaglandin E2 (PGE-2) levels, were extracted and subjected to meta-analysis.

Results Twenty-one studies involving 1705 patients were included. The studies were conducted across seven countries, and their sample sizes ranged from 24 to 160. Meta-analysis revealed that the levels of CRP (SMD = -0.906, 95% CI = -1.543 ~ -0.269, *P* value = 0.005) and TNF-alpha (SMD = -0.921, 95% CI = -1.817 ~ -0.026, *P* value = 0.044) were significantly lower in the curcumin group than in the placebo group. However, there were no significant differences in the ESR (SMD = -0.064, 95% CI = -0.064 ~ 0.541, *P* value = 0.836), IL-1beta (SMD = -0.362, 95% CI = -0.816 ~ 0.092, *P* value = 0.118), IL-6 level (SMD = -0.218, 95% CI = -0.806 ~ 0.370, *P* value = 0.467), or PGE-2 level (SMD = 0.413, 95% CI = -0.312 ~ 1.139, *P* value = 0.264) between the curcumin and control groups. The sensitivity test for CRP indicated that no individual study had a significant influence on the pooled results, while TNF-alpha was not as consistent as that of CRP. The visual analysis of the funnel plots revealed no publication bias.

Conclusion Curcumin led to a decrease in the serum CRP and TNF-alpha levels in patients with knee OA. Additional large-scale studies or even synovial fluid analyses are recommended to further evaluate the effects of curcumin on proinflammatory biomarkers in patients with knee OA.

*Correspondence:
Yi-Shiung Horng
yshorng2015@gmail.com
Full list of author information is available at the end of the article



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Keywords Curcumin, Knee osteoarthritis, Inflammatory biomarker, Systematic review and meta-analysis, C-reactive protein (CRP), Tumor necrosis factor-alpha (TNF-alpha), Interleukin-1 beta (IL-1beta), Interleukin 6 (IL-6), Erythrocyte sedimentation rate (ESR), Prostaglandin E2 (PGE-2)

Background

Knee osteoarthritis (OA) is a degenerative disease that strongly affects the activity and quality of life of elderly individuals. There were 595 million people suffering from knee OA in 2020, and the prevalence was greater than 5.5% in all world regions, ranging from 5.7% in Southeast Asia to 8.6% in the high-income Asia Pacific region [1]. Knee OA is characterized by articular cartilage loss, bone remodeling, and periarticular muscle weakness, resulting in joint swelling, deformity, painful sensations, and disability. Traditional pharmacological treatments with nonsteroidal anti-inflammatory drugs (NSAIDs) cause unfavorable side effects, such as gastric mucosal or small bowel injuries, cardiovascular disease, renal injury, and hepatotoxicity [2, 3]. Therefore, a considerable amount of research has been conducted to identify effective alternative therapies. Among them, herbal medicines such as *Curcuma longa* or *Boswellia serrata* extract have been reported to be effective alternative treatments for knee OA [4].

Curcuma longa has been used in traditional Chinese medicine as well as Ayurvedic medicine [5, 6]. Curcumin is a polyphenolic compound extracted from curcuma powder and is known to have anti-inflammatory [5], antioxidant, antiviral, and anticarcinogenic [7] effects. Furthermore, curcumin has been shown to improve glycemic control and lipid metabolism [8] and to enhance immunomodulatory activity [9]. In recent years, numerous studies have shown the beneficial effects of curcumin on musculoskeletal diseases, neurodegenerative diseases, cardiovascular diseases, metabolic diseases, diabetes mellitus, cancer and pain [10]. Curcumin has also been shown to modulate proinflammatory cytokines (e.g., tumor necrosis factor (TNF-alpha), interleukin (IL)-1beta, IL-6, and prostaglandin E2 (PGE-2)) [10, 11]. Interleukins are a large group of cytokines produced mainly by T cells, although some are also produced by mononuclear phagocytes or by tissue cells [12]. As a member of the interleukin family, IL1beta is strongly associated with cartilage destruction; IL-6 has been reported to be correlated with the severity of knee OA [13]; and TNF-alpha can drive the inflammatory cascade [14]. PGE-2 has been reported to exert an antianabolic effect on human adult articular cartilage and act as an angiogenesis stimulator in synovial fluid in a paracrine manner [15–17].

A previous systematic review and meta-analysis (SRMA) revealed that curcumin was an effective treatment for knee OA, as indicated by several patient-reported outcomes [10, 18–20], such as pain scale scores, the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and the Knee Injury and Osteoarthritis Outcome Score (KOOS). For example, Hsiao et al. reported that, compared with NSAIDs, curcuminoids are associated with greater pain relief in patients with knee OA [21]. However, no previous SRMA has examined the therapeutic effects of curcumin on serum inflammatory biomarkers in patients with knee OA, and previous RCTs on this topic have yielded inconsistent results. For example, the RCT by Rahimnia AR et al. revealed no significant differences between the curcumin group and the control group in terms of the serum levels of IL-4, IL-6, TNF-alpha, transforming growth factor-beta (TGF-beta), and high-sensitivity C-reactive protein (hsCRP), as well as the erythrocyte sedimentation rate (ESR) [22]. On the other hand, Shokri-Mashhadi et al. examined the findings of three RCTs and reported that curcumin led to significant decreases in C-reactive protein (CRP), IL-6, and IL-1beta levels as well as a significant decrease in the ESR [23]. Therefore, the aim of this SRMA was to examine the effects of curcumin on changes in serum inflammatory biomarkers in patients with knee OA.

Methods

This SRMA was conducted in accordance with the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 guidelines [24] shown in Supplement 1 and was registered in Prospero (CRD42024558196).

Eligibility criteria

Randomized controlled trials (RCTs) were considered eligible. The Population, Intervention, Comparator, and Outcome (PICO) framework was used to develop the following inclusion criteria: 1) P: adults with a diagnosis of knee OA or knee joint degenerative disease; I: any oral medication, including curcumin, regardless of any combination with other medications, such as NSAIDs; C: all control groups; O: serum biomarkers, including ESR, CRP, IL-1beta, IL-6, TNF-alpha, and PGE-2. The exclusion criteria were as follows: (1) animal models; (2)

lacked essential data; (3) lacked full text; or (4) were not written in English.

Search strategy and literature research

We searched the PubMed, MEDLINE, Embase, Web of Science, Cochrane Library, EBSCOhost, and Clinical-Trials.gov databases from inception to March 28, 2025. We used the Ovid platform to search the Embase and MEDLINE databases simultaneously. The keywords used to search the Web of Science databases were as follows: curcumin or turmeric or curcuminoid or "curcumin-glucuronoside" or curcuma or demethoxycurcumin or bisdemethoxycurcumin (All Fields) AND "knee osteoarthritis" or "knee OA" or "Degenerative joint disease of the knee" or "Osteoarthritis of the knee joint" or "Knee joint degeneration" or "Knee cartilage degeneration" or "Knee joint osteoarthritis" or "Knee joint arthrosis" or "Knee joint degenerative arthritis" or "Tibiofemoral osteoarthritis or Gonarthrosis" (All Fields) AND randomized or random (All Fields). The other search strategies are shown in Supplement 2. We also created an email alert on Embase to be notified about new publications.

Data extraction

Han-Chien Hsueh and Guan-Ru Ho independently performed the literature screening process on the basis of the inclusion and exclusion criteria. The literature search was first conducted in the Embase database, and then, the search terms and Medical Subject Heading (MeSH) terms were adapted to other databases. The results were summarized by Han-Chien, and all the data were imported into EndNote (version 21). Any disagreements were resolved through discussion with Yi-Shiung, Horng. The following data were extracted from the included studies: title, year of publication, first author, country, treatment therapy, curcumin dosage and type, number of interventions and control groups, treatment duration, and serum laboratory data (including baseline and post-treatment data).

Quality assessment

The revised Cochrane risk-of-bias tool for randomized trials (RoB 2) [25] was used to assess the risk of bias. Each study was judged as having a low, unclear, or high risk of bias in the following domains: selection of participants, classification of interventions, deviations from intended interventions, missing data, measurement of outcomes, and selective reporting of the result. We also used the grading of recommendations, assessment, development, and evaluation (GRADE) framework to assess the certainty of the outcomes, which was categorized into four levels: "high", "moderate", "low", or "very low".

Data analysis

The effect of curcumin was determined by examining changes in serum biomarkers between pretherapy and posttherapy. Review Manager 5.4.1 and Comprehensive Meta-Analysis (CMA) version 4 software were used for the statistical analyses. Standardized mean differences (SMDs) and their corresponding 95% confidence intervals (CIs) were used for effect analysis. Statistical heterogeneity was assessed via both the Q test and I-squared values (I^2). The Q test was used to test the statistical significance of heterogeneity. I^2 and 95% CIs were calculated to evaluate the extent of heterogeneity between different research results [26]. According to the Cochrane guidelines, an I^2 statistic of 0% to 40% might not be significant, an I^2 statistic of 30% to 60% may indicate moderate heterogeneity, an I^2 statistic of 50% to 90% may indicate substantial heterogeneity, and an I^2 statistic of 75% to 100% may represent considerable heterogeneity [27]. Pooled meta-analysis was conducted via a random effects model on the basis of the degree of statistical heterogeneity across studies. CMA version 4 was used to construct forest plots and perform sensitivity analysis, and Review Manager 5.4.1 software was used to assess the risk of bias. Sensitivity analysis was performed via the leave-one-out method to evaluate the impact of each individual study on the pooled results. To analyze the risk of publication bias, funnel plots were constructed, and Egger's test was performed for each outcome. Meta-regression was performed with age, sex, BMI, and duration of treatment as covariates.

Results

Search results

The study selection process is illustrated in Fig. 1. The literature search initially yielded 388 studies. After approximately 172 duplicate records were excluded, 216 studies remained for screening. After screening the abstracts and titles, we excluded 109 records because they were not relevant. The full texts of the remaining 107 papers were screened, and 80 were excluded because they did not meet the inclusion criteria. Therefore, we included 27 studies. The study of G. Belcaro et al. was excluded due to missing standard deviation (SD) data [28]. The study by Z. Wang et al. lacked full content [29]. Two RCTs lacked the specific inflammatory biomarkers we were focusing on [30, 31]. The study by Y. Henrotin et al. showed baseline CRP data but lacked post-therapy data [32]. The study by L. Calderón-Pérez et al. expressed the data in log form, which could not be converted to general figures; therefore, these data were excluded [33]. A total of 21 RCTs were ultimately included in this meta-analysis [9, 22, 34–52].

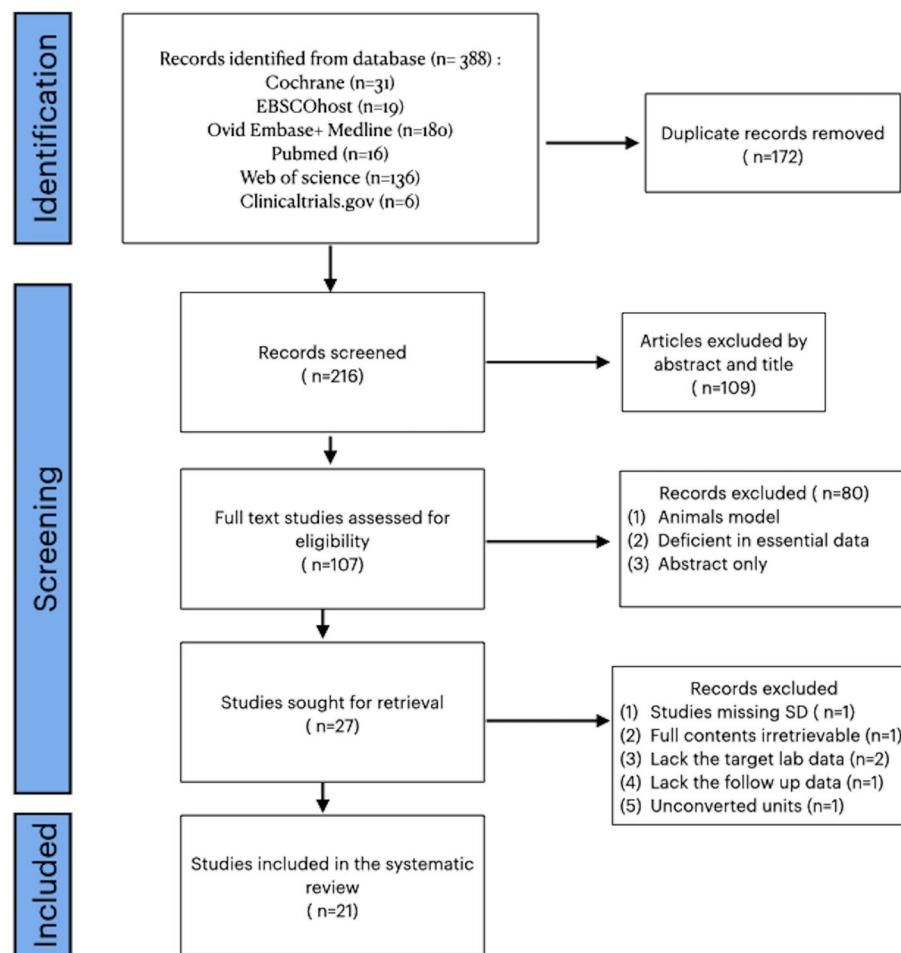


Fig. 1 Flowchart of the study selection process

Characteristics of the included studies

The characteristics of the included studies are presented in Table 1. The included studies were published from 2004 to 2025, and their sample sizes ranged from 24 to 160. The total number of participants was 1705, including 852 in the intervention group and 853 in the control group. The mean ages of the participants ranged from 48 to 71 years. The control treatments in these trials were as follows: 11 studies using placebo only; 7 studies using NSAIDs; 1 study using paracetamol and 2 studies using glucosamine plus chondroitin. Among these trials, 13 studies used curcumin-based medication as the intervention, 6 studies used a curcumin formula, and 2 studies used curcumin + NSAID. The curcumin formulas used included Acujoint, Instaflex joint support supplement, CartiJoint Forte, and RA-11; additionally, two studies used NXT15906 F6. Two studies included patients treated with curcumin plus NSAIDs. The dosage of curcumin ranged from 80

mg/day to 1500 mg/day. The duration of the intervention among the included trials ranged from 1 week to 4 months.

Quality assessment

The results of the risk of bias assessment via the RoB2 tool are shown in Fig. 2. The vast majority of the studies had a low risk of bias regarding random sequence generation, but four RCTs did not provide a description of the random sequence generation process. In terms of allocation concealment bias, six studies had an unclear or high risk of bias. Eight studies were graded as having a low risk of bias for the blinding of participants and personnel, whereas five studies were considered to have a high risk of bias, and three studies that did not mention a clear description of blinding were considered to have an unclear risk of bias. Regarding the blinding outcome assessment, two studies had an unclear risk of bias, and two studies had a high risk of bias. Using the GRADE

Table 1 Characteristics of twenty-one studies

Study	Country	Sample size		Age		Grade	Formulation, dose		Duration	Lab data
		Intervention	Control	Intervention	Control		Experimental intervention	Control		
Chopra, 2004 [34]	India	45	45	(mean age 59 years, range 35–75; 35 females)	mean age 55 years, range 42–78; 32 females	ACR criteria: clinical plus radiologic criteria (osteophyte)	RA-11 (Withania somnifera, Boswellia serrata, Zingiber officinale, and Curcuma longa)	Placebo	Week 16, Month 4	ESR, Hb
Nieman, 2013 [35]	USA	50	51	57.6 ± 0.9 (N = 49; 41 F, 8 M)	58.3 ± 0.8 (N = 51; 42 F, 9 M)	Nil (Self-reported history (> 3 months) of joint pain in the knees, hip, ankles, shoulders, or hands. Minimum symptom severity was ensured by using a WOMAC pain index score of at least 2 points)	Instaflex™ Joint Support supplement (Instaflex) (Direct Digital, Charlotte, NC): glucosamine sulfate, methylsulfonylmethane, white willow bark extract, ginger root concentrate, Boswellia serrata extract, turmeric root extract, cayenne, and hyaluronic acid	Placebo	Week 8, Month 2	IL-6, CRP, TNF-α
Rahimnia, 2015 [22]	Tehran, Iran	27	26	57.32 ± 8.78	57.57 ± 9.05	ACR criteria	Curcuminoids (500 mg curcuminoids + 5 mg Bioperine®) (C3 complex®); 1 500 mg/day;	Placebo	Week 6	IL-4, IL-6, TGF-β (pg/ml) TNF-α, Hs-CRP, ESR, Leukocytes, Lymphocytes, Polymorphonuclears, Platelets
(Srivastava et al., 2016) [36]	India	78	82	50.27 ± 8.63	50.23 ± 8.08	ACR criteria K&L grading scale I ~ IV	CL extract 500 mg + diclofenac 50 mg/day	500 mg placebo + diclofenac 50 mg/day	Day 60, Month 2	IL-1b (pg/ml) ROS (MFI) MDA (nmol/ml)
Sterzi, 2016 [37]	Italy	23	27	71.3 ± 8.8	71.0 ± 8.1	ACR criteria K&L grading scale II ~ III	Two tablets of CartiJoint Forte per day,	two placebo tablets per day	Week 8, Month 2	CRP ESR
(Harovan et al., 2018) [38]	Armenia	57	54	54.65 ± 8.84	56.04 ± 8.55	K&L grading scale I ~ III	Curamed contains 552–578 mg of BCM-95® as a dry extract, corresponding to 500 mg curcuminoids	Placebo	Week 12, Month 3	ESR, mm/h Reference range: 2–20 mm/h CRP, mg/L Reference range: < 5 mg/L

Table 1 (continued)

Study	Country	Sample size		Age		Grade	Formulation, dose		Duration	Lab data
		Intervention	Control	Intervention	Control		Experimental intervention	Control		
(Panda et al., 2018) [39]	India	25	25	55.20 ± 8.58	53.12 ± 8.25	ACR criteria K&L grading scale II ~ III	Curcure: one capsule of 500 mg once daily	placebo	Day 60, Month 2	CBC/DC ESR Random glucose Cr/BUN, Bilirubin, GOT, GPT, Alk-P, Na, K, Albumin Urine PH, gravity
(Gupte et al., 2019) [41]	India	17	25	57 ± 7.5	54 ± 8	ACR criteria: clinical plus radiologic criteria (osteo-phyte)	Capsules twice daily (80 mg curcumin/capsule)	Ibuprofen (400 mg/day) + Placebo (dextrin)	Day 90, Month 3	PGE2, LTb4 IL-6, IL-1β TNF-α, CTX-II
Shep, 2019 [42]	India	70	69	53.09 ± 4.17	52.14 ± 3.76	ACR criteria: clinical plus radiologic criteria (osteo-phyte)	curcumin (BCM-95®) 500 mg three times daily	diclofenac 50 mg tablet two times daily	Month 1	CBC/DC Peripheral smear, ESR
Amalraj, 2019 [40]	india	12	12	53.2 ± 10.42	52.8 ± 10.36	ACR criteria: clinical plus radiologic criteria (osteo-phyte)	Acujo TM (Boswellia serrata, Piper nigrum, and Kaempferia galanga and bioavailable form of Curcuma longa—Cureit TM) 250 mg of Acujo TM /day	Glucosamine (1500 mg) and Chondroitin (1200 mg)	Month 3	Leucocyte count ESR, hs-CRP
(Atabaki et al., 2020) [9]	Iran	15	15	49.13 ± 1.50	48.26 ± 1.32	ACR criteria K&L grading scale II ~ III	SinaCurcumin [®] (80 mg of the curcumin) + 50 mg diclofenac/day	placebo + 50 mg diclofenac/day	3 month	CRP (mg/l) ESR (mm)
Heidari-Beni, 2020 [43]	Iran	30	30	49.93 ± 10.88	49.90 ± 11.32	K&L grading scale II ~ III	Curcumin (300 mg), gingerols (7.5 mg), and piperine (3.75 mg), twice a day	Naproxen	4 weeks	PGE2
Shep, 2020 [44]	India	71	69	52.55 ± 4.46	52.14 ± 3.76	ACR criteria: clinical plus radiologic criteria (osteo-phyte)	Curcuminoid complex 500 mg (BCM-95) with diclofenac 50 mg 2 times daily	Naproxen 250 mg capsules twice/day	Month 1	Platelets, ESR, Hb, RBC, WBC
Singhal, 2021 [46]	India	73	71	53.1 (10.9)	50.8 (9.9)	ACR criteria K&L grading scale II ~ IV	Turneric extract 1000 mg (one capsule of 500 mg BCM-95 [®] ; twice daily)	Paracetamol 650 mg thrice a day	Week 6, Month 1.5	CRP TNF-α

Table 1 (continued)

Study	Country	Sample size		Age		Grade	Formulation, dose		Duration	Lab data
		Intervention	Control	Intervention	Control		Experimental intervention	Control		
Thomas, 2021 [58]	India	35	37	51.7 ± 5.52	52.3 ± 4.59	K&L grading scale I ~ III	CGM(126.2 mg curcumin, 23.6 mg demethoxy-curcumin, and 4.3 mg bisdemethoxycurcumin): 400 mg/day	GLN/CHN: 1,000 mg glucosamine hydrochloride (GLN) and 830 mg chondroitin sulphate (CHN) per day	Week 6. Month 1.5	GOT,GPT,Cr, BUN, TC, TG, LDL cholesterol, HDL cholesterol, VLDL, CBC, IL-1, IL-6, sVCAM, hs-CRP
Nasir, 2021 [45]	Iraq	23	20	56.70 ± 6.42	54.26 ± 8.42	KOA confirmed by radiographic analysis	Curcumin 500 mg and Piperine 5 mg/day	placebo	Month 3	CRP
Maiss S. Baqer1, 2022 [49]	Iraq	21	21	B: 50.52 ± 9.90	A: 49.52 ± 8.07	K&L grading scale II ~ III	meloxicam 15 mg + Curcumin 800 mg 2 caps/day	A: meloxicam 15 mg once daily	Month 3	IL-1β, IL-6, TNF-α
Kare, 2022 [48]	india	30	30	54.3 ± 7.8	53.3 ± 9.3	ACR criteria K&L grading scale II ~ III	250 mg of NXT15906 F6 (T. indica seeds and C. longa rhizome extracts.)	placebo	Day 56, week 8 Month 2	TNF- alpha CRP IL-6
Wang, 2023 [51]	Australia	34	31	61.1 (8.7)	61.6 (8.7)	ACR criteria Ultrasonography defined effusion-synovitis (≥ 4 mm effusion depth	CL:1000 mg/day	placebo	Week 12. Month 3	MMP-3, IL6, hsCRP, TNFα
Prasad, 2023 [50]	India	46	44	53.1 ± 9.09 (40–70)	53.4 ± 9.26 (40–70)	ACR criteria K&L grading scale II	NXT15906 F6 (T. indica seeds and C. longa rhizome extracts): 250 mg/day	Placebo	Month 1	CRP
Thanawala, 2025 [52]	india	70	69	57.03 (6.79)	55.67 (6.91)	National Institute for Health and Care Excellence [NICE] criteria K&L grading scale II ~ III	WDTE60 N (single daily low dose of 250 mg, containing 150 mg of curcuminoids)	placebo	Month 3	hsCRP, TNF-α, IL-6, and IL-1β

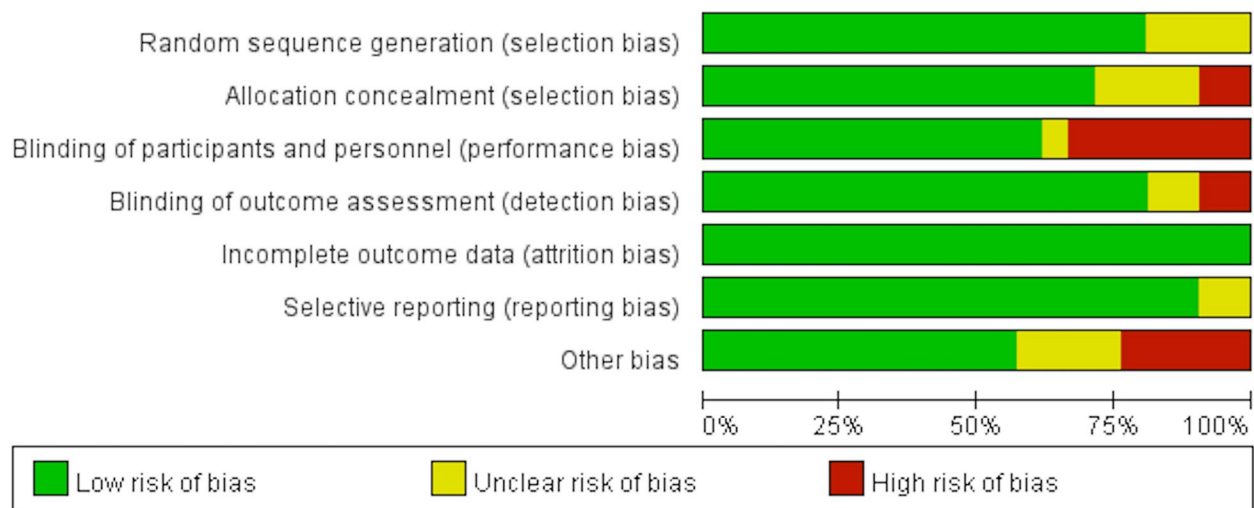


Fig. 2 Risk of bias assessment for the 21 included RCTs: (A) graph; (B) summary

approach, the quality of the studies was evaluated, and the results are presented in Table 2. Most studies were downgraded to low or very low quality. Concerning inconsistency, significant heterogeneity was observed across indicators, resulting in a downgrade in quality. For imprecision, several indices crossed the threshold of no effect or exhibited wide confidence intervals. No concerns were noted regarding indirectness.

Laboratory results

Inflammatory biomarkers

Thirteen studies reported pre- and post-treatment CRP levels, and 9 studies reported the ESR. Figure 3 shows the forest plot of the SMD and 95% CI for CRP levels. The meta-analysis revealed a significant difference in the pooled SMD of CRP levels between the curcumin group and the control group (SMD = -0.906, 95% CI = -1.543 ~ -0.269, P value = 0.005), with considerable heterogeneity ($I^2 = 94.84\%$). However, no significant difference was observed in the ESR between the curcumin and control groups, as shown in Fig. 4 (SMD = -0.064, 95% CI = -0.670 ~ 0.541, P value = 0.836), with considerable heterogeneity ($I^2 = 92.6\%$).

Proinflammatory cytokines

Four proinflammatory cytokines, including IL-1beta, IL-6, PGE2, and TNF-alpha, were analyzed. Only TNF-alpha significantly differed between the curcumin and control groups. Five studies examined IL-1beta, and the meta-analysis revealed no significant between-group differences in this cytokine (Fig. 5, SMD = -0.362, 95% CI = -0.816 ~ 0.092, P value = 0.118), with substantial heterogeneity ($I^2 = 80.35\%$). IL-6 was examined in eight

studies, and the meta-analysis revealed no significant between-group differences in this cytokine, as shown in Fig. 6 (SMD = -0.218, 95% CI = -0.806 ~ 0.370, P value = 0.467), with considerable heterogeneity ($I^2 = 90.67\%$). Eight studies examined TNF-alpha; the meta-analysis revealed significant between-group differences in this cytokine, as shown in Fig. 7 (SMD = -0.921, 95% CI = -1.817 ~ 0.026, P value = 0.044), with considerable heterogeneity ($I^2 = 95.85\%$). Only two studies have examined serum PGE-2 levels; the meta-analysis revealed no significant between-group differences in this cytokine, as shown in Fig. 8 (SMD = 0.413, 95% CI = -0.312 ~ 1.139; P value = 0.264), with substantial heterogeneity ($I^2 = 68.5\%$).

Meta-regression

Owing to the high heterogeneity of the results, we performed meta-regression to verify the confidence of our two significant outcomes: CRP and TNF-alpha. We selected age, sex, BMI, and duration of treatment as covariates of the meta-regression. The results of the random effects multiple meta-regression models are shown in Table 3. All of the P values were greater than 0.05 showing no contribution to the heterogeneity.

Sensitivity analysis

We conducted sensitivity analyses for CRP and TNF-alpha since they were the only two biomarkers that significantly differed between the curcumin and control groups. Using the leave-one-out method, each trial was excluded iteratively, and a meta-analysis was performed on the remaining studies. As shown in Fig. 9, no individual study had a significant influence on the

Table 2 Grading of recommendations assessment, development, and evaluation evidence profile for the role of curcuminoids in knee osteoarthritis of the twenty-one studies included in the meta-analysis

Certainty assessment							№ of patients		Effect	Certainty	Importance	
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	curcumin	placebo	Relative (95% CI)	Absolute (95% CI)		
C reactive protein (follow-up: range 1 weeks to 4 months; assessed with: standardized mean difference)												
13	randomised trials	not serious ^a	serious ^b	not serious	serious ^b	none	782	784	-	SMD 0.907 SD lower (1.633 lower to 0.181 lower)	⊕⊕○○ Low	IMPORTANT
Erythrocyte sedimentation rate (follow-up: range 6 weeks to 16 weeks)												
9	randomised trials	serious	serious ^c	not serious	not serious	none	345	342	-	SMD 0.064 SD lower (0.67 lower to 0.541 higher)	⊕⊕○○ Low	
Prostaglandin E (follow-up: range 4 weeks to 3 months; assessed with: standardized mean difference)												
2	randomised trials	serious	serious ^d	not serious	serious ^e	none	47	55	-	SMD 0.413 SD higher (0.312 lower to 1.139 higher)	⊕○○○ Very low	
Tumor necrosis factor-alpha (follow-up: range 6 weeks to 90 days; assessed with: standardized mean difference)												
8	randomised trials	not serious	serious ^f	not serious	serious ^b	none	301	303	-	SMD 0.921 SD lower (1.817 lower to 0.026 lower)	⊕⊕○○ Low	IMPORTANT
Interleukin-6 (follow-up: range 42 days to 90 days; assessed with: standardized mean difference)												
8	randomised trials	serious	serious	not serious	serious ^e	none	284	290	-	SMD 0.218 SD fewer (0.806 fewer to 0.37 more)	⊕○○○ Very low	
Interleukin-1 beta (follow-up: range 42 days to 90 days; assessed with: standardized mean difference)												
5	randomised trials	serious	serious ^g	not serious	serious ^e	none	221	234	-	SMD 0.362 SD fewer (0.816 fewer to 0.092 more)	⊕○○○ Very low	

CI confidence interval, SMD standardised mean difference

For CRP, the MD was -2.840 (95% CI: -4.151 to -1.529), indicating a significant effect of curcumin in reducing CRP levels in patients with knee osteoarthritis (KOA). Considering that the normal CRP range is < 1 mg/L, a reduction of 1.5 to 4.1 mg/L suggests that curcumin may have a meaningful anti-inflammatory effect, potentially bringing elevated CRP levels back into the normal range. Nonetheless, due to a conservative approach toward interpreting statistical significance, we rated the imprecision for CRP as serious

Regarding TNF-α, the normal range is considered to be <8.1 pg/mL, and the MD observed was -3.608 (95% CI: -4.731 to -2.485). Although this result is statistically significant, the clinical impact may be limited. Therefore, we also rated the imprecision for TNF-α as serious

^aTwo studies had performance bias due to incomplete blinding of participants and personnel, and three studies had selection bias due to vague description of random sequence generation

^bStatistically significant of heterogeneity with low *p* value and high *I* square

^c*I*2 showed 92.6%, indicating considerable heterogeneity

^d*I*2 showed 68.5%, indicating substantial heterogeneity

^edue to low sample size < 400

^f*I*2 showed 95.7%, indicating considerable heterogeneity

^g*I*2 showed 89.9%, indicating considerable heterogeneity

^hDue to the statistically significant results for CRP and TNF-α, we searched previous studies for the minimally clinically important difference (MCID) of these two biomarkers. However, we were unable to find any relevant data. Therefore, we attempted to analyze the mean difference (MD) in CRP and hsCRP as subgroups to interpret their clinical relevance

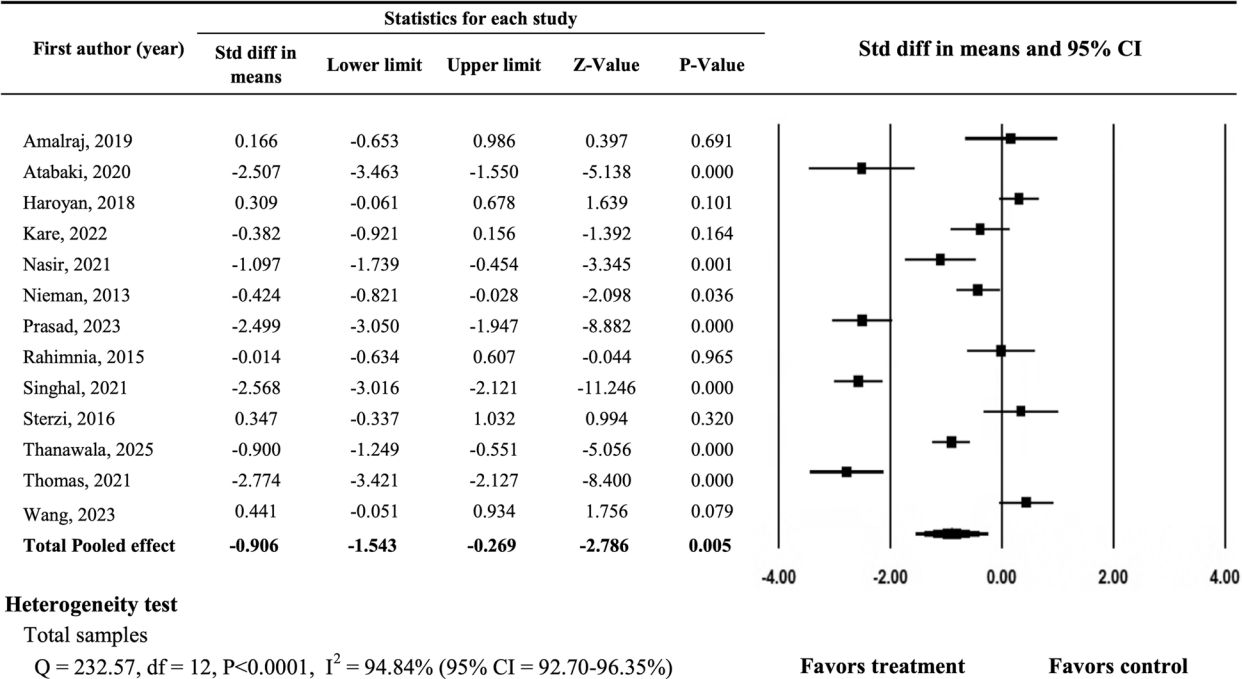


Fig. 3 Forest plot showing the standard mean differences and 95% confidence intervals for C-reactive protein

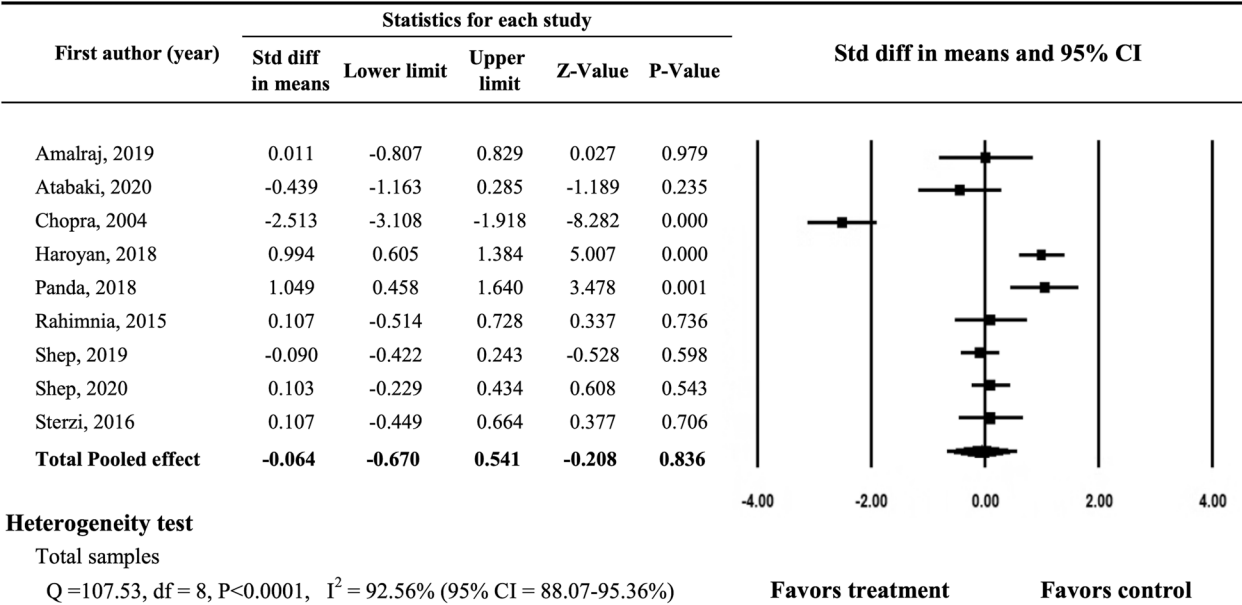


Fig.4 Forest plot showing the standardized mean differences and 95% confidence intervals for the ESR

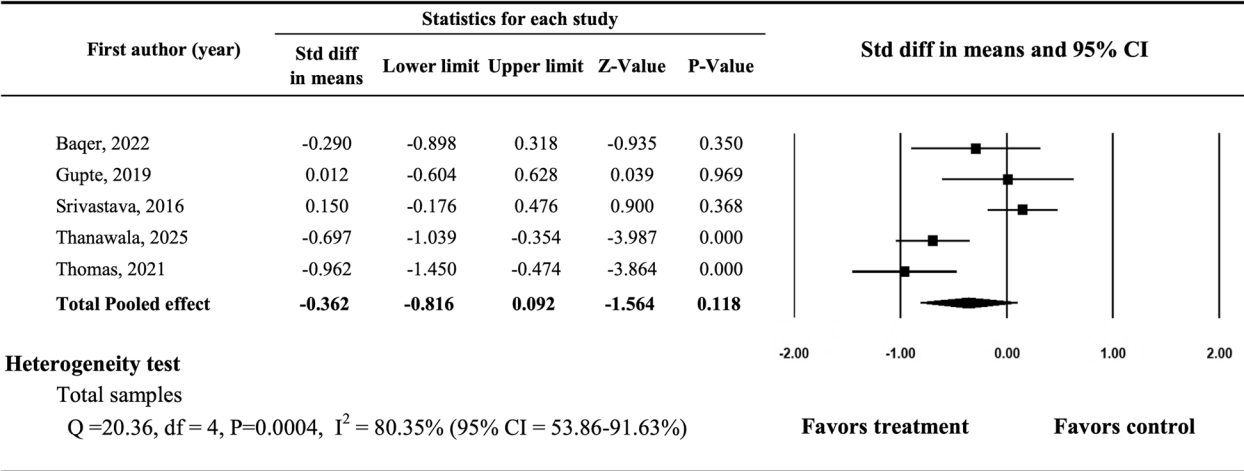


Fig. 5 Forest plot showing the standardized mean differences and 95% confidence intervals for IL-1

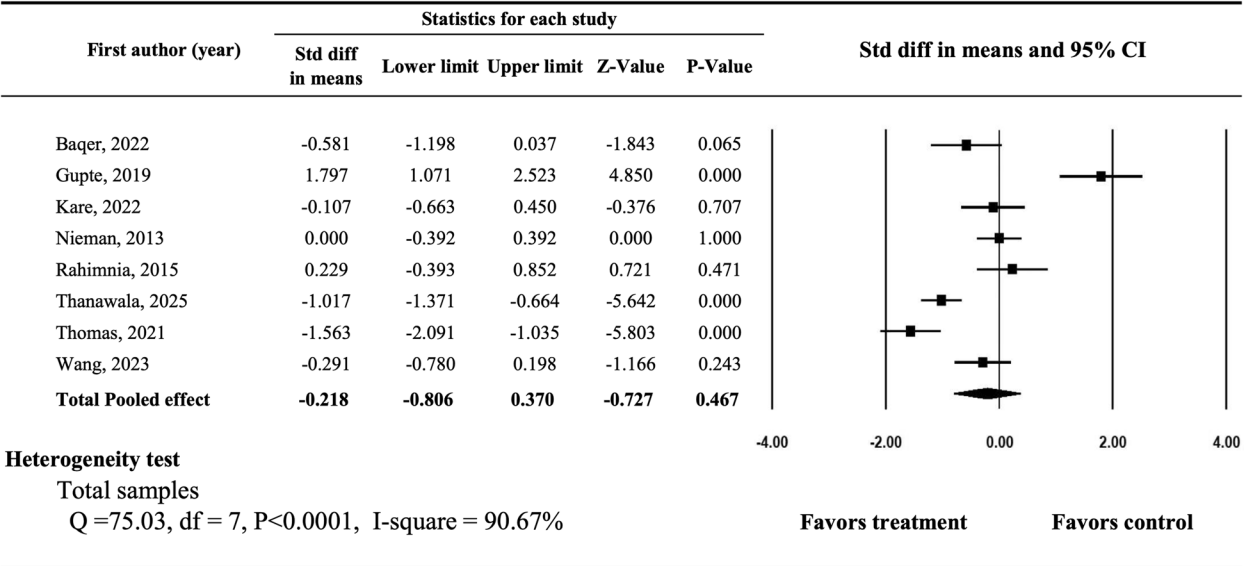


Fig. 6 Forest plot showing the standardized mean differences and 95% confidence intervals for IL-6

overall pooled results of CRP, thus indicating that the findings are robust. However, the sensitivity analysis of TNF-alpha shown in Fig. 10 was not as consistent as that of CRP. Therefore, the results of TNF-alpha should be interpreted with caution.

Publication bias

Funnel plots were constructed for both inflammatory and proinflammatory biomarkers (Supplement 3). The results of Egger’s test were not significant for any of the serum inflammatory biomarkers, thus indicating that there was no publication bias.

Discussion

This SRMA of 21 RCTs revealed that curcumin was more effective than placebos in terms of decreasing serum CRP and TNF-alpha levels but not in terms of changing the ESR or the serum levels of IL-1beta, IL-6, and PGE-2. CRP is a direct indicator of the inflammatory response, which is primarily induced by the IL-6 activity on the gene that is responsible for the transcription of CRP during the process of inflammation [53]. The ESR is an indirect measure of the level of inflammation in the body. Our results showed that the CRP level was more

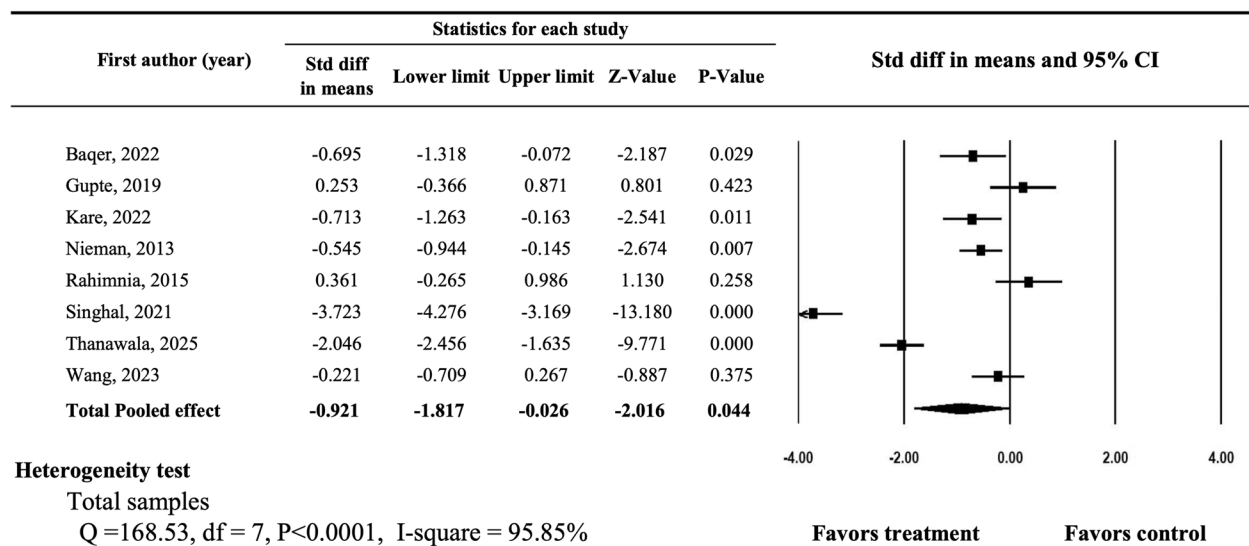


Fig. 7 Forest plot showing the standardized mean differences and 95% confidence intervals for TNF-α

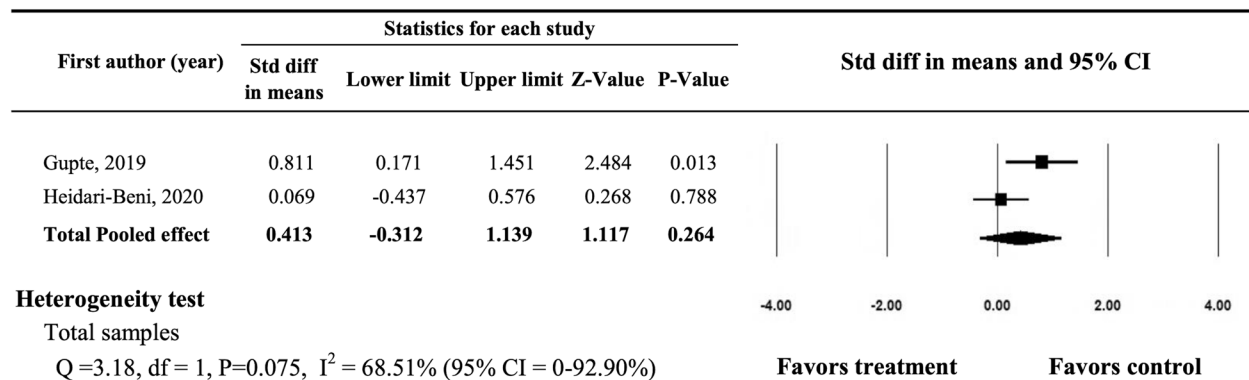


Fig. 8 Forest plot showing the standardized mean differences and 95% confidence intervals for PGE2

Table 3 Results of random effects multiple meta-regression models

Outcomes	Covariates	β	SE	95% CI	Z	p-value
CRP	Age	0.049	0.106	-0.159	0.256	0.645
	Female(%)	-1.369	1.863	-5.020	2.282	0.462
	BMI	0.150	0.169	-0.182	0.481	0.377
	Duration (weeks)	0.177	0.099	-0.018	0.372	0.075
TNF-alpha	Age (years)	0.174	0.178	-0.175	0.523	0.329
	Female (%)	5.312	5.847	-6.149	16.772	0.364
	BMI	-0.246	0.401	-1.031	0.540	0.540
	Duration (weeks)	0.188	0.166	-0.136	0.513	0.256

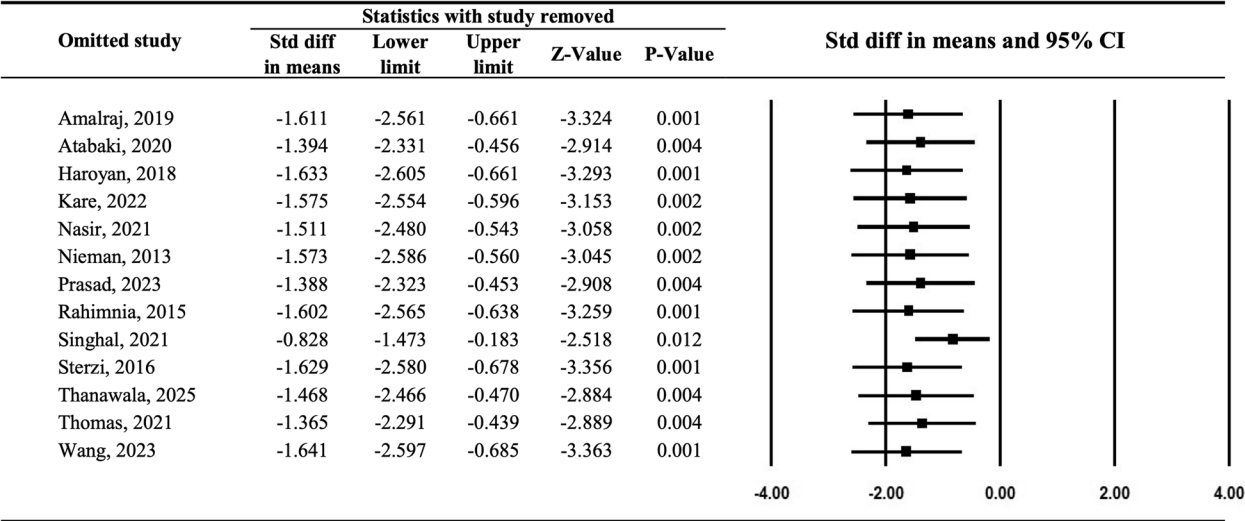


Fig. 9 Forest plot for the sensitivity analysis of C-reactive protein

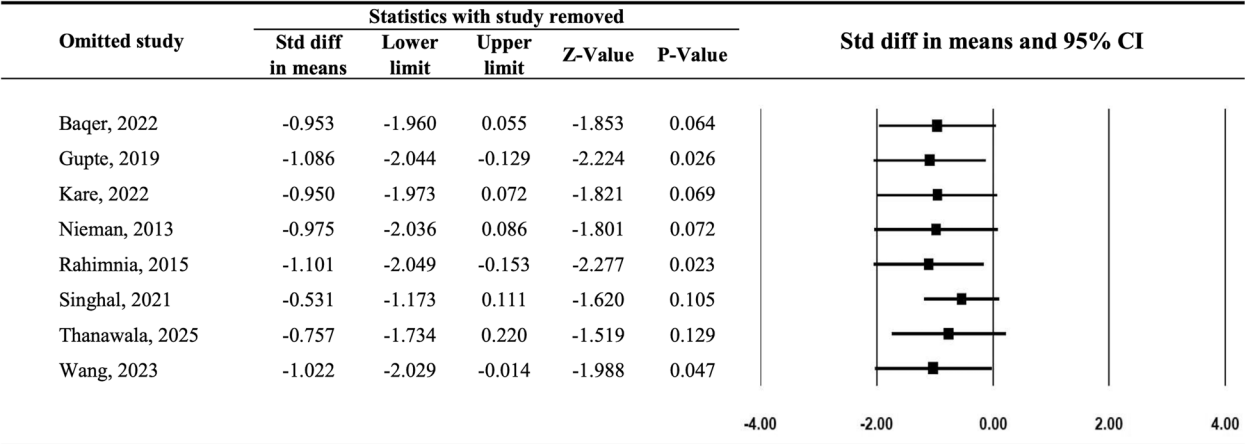


Fig. 10 Forest plot for the sensitivity analysis of TNF-alpha

sensitive than the ESR in terms of responding to the anti-inflammatory effects of curcumin. This finding is consistent with the results of a 5-year cohort study in which the serum CRP level was found to be associated with changes in knee pain [54].

We examined IL-1beta, IL-6, TNF-alpha and PGE-2 because IL-1beta, IL-6 and TNF-alpha are the three most important proinflammatory cytokines in knee OA [55]. Although a previous systematic review of four RCTs revealed no between-group differences in TNF-alpha, TNF-beta, or IL-6 levels [56], our meta-analysis revealed a borderline significant difference in TNF- α levels ($p = 0.044$) between the curcumin and control groups. TNF- α can induce chondrocytes to synthesize prostaglandin E2 and nitric oxide, which might exacerbate local

inflammation and pain intensity [57]. IL-1beta drives synovitis and acts as a potent instigator of cartilage degradation in OA via matrix metalloproteinase-3 (MMP-3) and MMP-13 induction in chondrocytes. IL-6 is also an essential marker for cartilage loss in OA that can provide useful information for the prediction of disease outcomes, especially in obese and older individuals [58]. IL-1 β can suppress the synthesis of type II collagen and aggrecan, the key constituents of cartilage [41]. A previous study revealed that elevated serum concentrations of IL-6 and TNF- α were associated with cartilage degradation in knee OA [55]. Our results also revealed that the serum levels of IL-1beta, IL-6, and PGE-2 decreased in the curcumin group, although the differences did not reach statistical significance, as reported previously by

Gupte [41]. Therefore, the effects of curcumin on pro-inflammatory cytokines still need further large-scale investigations.

The pathophysiology of knee OA can be divided into three main steps. First, long-term overuse or overloading of the knee joint (e.g., in obese individuals) could lead to surface fibrillation of the cartilage and irregularity of the knee joint [57]. Second, after the cartilage surface is damaged, chondrocytes undergo mitosis and clustering to renew the chondrocytes; however, the catabolic activity is greater than the synthetic activity. Consequently, the collagen matrix is damaged as well, which could cause irreversible matrix degradation [59]. Moreover, TNF-alpha and IL-1 stimulate chondrocytes to produce matrix metalloproteinases (MMPs) and plasminogen activators, which degrade matrix proteoglycans and collagen [16]. Moreover, remodeling of subchondral bone stimulates neovascularization. These vascular channels might interrupt the hypoxic environment of chondrocyte cells and facilitate biochemical communication, which exposes chondrocyte cells to cytokines and chemokines [2]. Third, this vicious cycle results in the continuous loss of cartilage and the formation of sclerosis, subchondral cysts, and osteophytes. Finally, the above nociceptive stimulation leads to progressive OA.

In an in vitro study of knee OA chondrocytes, curcumin significantly reduced IL-6, IL-8, and PGE-2 levels and decreased the MMP-3/tissue inhibitor of metalloproteinase-1 (TIMP-1) ratio by inhibiting MMP-3 synthesis [11]. Another in vitro study revealed a significant reduction in IL-1beta-stimulated PG and PGE 2 in explants [60]. This mechanism might correspond to the aforementioned second-step pathophysiological mechanisms in humans. However, a systematic review conducted by Dainese, P., et al. revealed conflicting evidence for associations between IL-6, TNF-alpha and knee pain in patients with knee OA [61]. Our meta-analysis also revealed no significant difference between the treatment and control groups in terms of changes in IL-1 and IL-6 levels. Therefore, the success observed in in vitro experiments could not be replicated in serum biomarkers in humans, but it might be reproducible in synovial fluid, as in vitro experiments might mimic the paracrine signaling property [15]. Inflammatory cytokines, including IL-1 β and IL-6, are produced by the synovium and chondrocytes and are expressed in the synovial fluid [49]. These cytokines increase the production of prostaglandin E2 (PGE2) via cyclooxygenase activation, resulting in further articular inflammation and the development of pain. In addition, in a rat model of KOA, the levels of IL-1beta, IL-6, and TNF-alpha in the synovial fluid of the KOA group were significantly increased, probably due to the local

inflammatory response of chondrocytes [62]. Therefore, further studies might evaluate the effects of curcumin on KOA by investigating changes in proinflammatory biomarkers in the synovial fluid of the knee joint.

In addition to its effects on knee OA, curcumin has also been reported to exhibit antioxidative and anti-inflammatory properties in patients with rheumatoid arthritis (RA). A previous RCT revealed a significant improvement in the CRP level exclusively in the curcumin group among RA patients [63]. Previous SRMAs of patients with RA have also demonstrated significant reductions in pain levels, rheumatoid factor levels, and serum biomarkers such as CRP and the ESR [64]. Additionally, Kondo, N. reported that the central pathogenesis of RA involves mainly TNF-alpha and IL-6 [65]. However, our study revealed only significant decreases in CRP and borderline significance in TNF-alpha levels, with no changes in the other proinflammatory cytokines. This discrepancy might be due to differences in the disease mechanisms between OA and RA. OA is a degenerative disease, whereas RA is an inflammatory autoimmune disease triggered by immune disturbances that produce numerous proinflammatory cytokines [66]. This tentative explanation is consistent with the findings of Sipe, J.D. [67].

Two previous systematic reviews examining curcumin and knee OA [23, 56] reported laboratory parameters as outcomes, including TNF-alpha, TNF-beta, IL-1beta, IL-4, IL-6, ESR, hs-CRP, Coll-2 and CTX II, oxidative stress malondialdehyde (MDA), reactive oxygen species (ROS), and superoxide dismutase (SOD) levels. In the review by Shokri-Mashhadi, N., et al., five studies reported inflammatory biomarkers [23]. Among these five studies, three showed significantly decreased levels of CRP, ESR, IL-6, and IL-1beta, whereas the other two studies did not observe changes in any biomarkers. However, another systematic review by Wang, Z., et al. reported that there were no significant between-group differences in TNF-alpha, TNF-beta, IL-6, and hs-CRP [56]. These two systematic reviews did not include meta-analyses of the reported inflammatory biomarkers. Our study is the first meta-analysis to examine the effects of curcumin on serum inflammatory biomarkers in patients with knee OA. Although the sample sizes for some of the biomarkers were small, our study demonstrated that, compared with the control treatments, curcumin led to greater improvements in serum CRP levels. The decrease in CRP might be due to the interruption of the vicious cycle of local inflammation. The decrease in oxidative stress indicated the cessation of chondrocyte apoptosis, so the use of CRP as a biomarker for testing the anti-inflammatory effects of curcumin might be promising. Larger studies for more accurate outcomes are recommended.

Most of the included RCTs were rated as having a low risk of bias. However, nine RCTs had an unclear or high risk of selection bias, and six RCTs had a high risk of bias in multiple categories. The sensitivity test for CRP indicated that no individual study had a significant influence on the pooled outcomes, thus suggesting that the results were robust. However, TNF-alpha was not consistent with CRP. Furthermore, there was no publication bias for any of the serum inflammatory biomarkers.

Strengths and Limitations

This was the first meta-analysis to evaluate the effects of curcumin on serum inflammatory biomarkers in patients with knee OA. We searched for the most recent studies as of March 2025. Most of the included studies had a low risk of bias and good quality. However, this study has several limitations. First, high heterogeneity was observed on the basis of the forest plot analysis, which might be due to the following three reasons: 1) high variation in the curcumin regimens of the intervention group; 2) different control groups (e.g., those with NSAIDs or not); and 3) different grades of KOA. Among the 21 included articles, eleven studies used curcumin alone for the treatment group; six studies used a curcumin formula that contained other ingredients, such as *Boswellia serrata* or glucosamine sulfate [34, 35, 37, 40, 48, 50]; and the other four studies used curcumin plus NSAIDs as the intervention [9, 36, 44, 48]. In addition to the high variability of the intervention group, the control group of seven papers used NSAIDs instead of placebo, which might be another reason for the increased heterogeneity. Considering the grading of KOA mentioned in the inclusion criteria, most of the studies ($n = 10$) were classified as grade I~III according to the Kallgren and Lawrence scale (K&L scale). The other studies used the American College of Rheumatology (ACR) criteria as a diagnostic tool (Supplement 4). With respect to the second limitation, the number of included studies for each outcome was small. Among the six inflammatory biomarkers examined herein, only CRP was reported in 13 articles; the other five biomarkers were reported in fewer than 11 studies. Third, since curcumin is lipophilic, different techniques were used to improve its bioavailability in 12 of the 21 included articles. For example, two studies [22, 45] added piperine, which could increase the bioavailability of curcumin by 2000% [68]. Three RCTs used BCM-95 as an intervention regimen and blended purified curcuminoids with turmeric volatile oil to increase its bioavailability [37, 38, 46]. One RCT used a nanotechnology to address the nanomicelle SinaCurcumin, which showed approximately 100% efficacy [9]. Under such high-variety circumstances, it is difficult to make direct

comparisons of the dosage and total amount of curcumin taken between studies.

Conclusion

The results of our meta-analysis revealed the anti-inflammatory effect of curcumin. Specifically, curcumin was found to significantly decrease serum CRP and TNF-alpha levels. However, the effects of curcumin on other serum inflammatory biomarkers need to be explored in additional larger and high-quality studies. Given that the mechanism of knee OA is complicated and multifactorial, further research on inflammatory biomarkers in synovial fluid might be important.

Abbreviations

OA	Osteoarthritis
SRMA	Systematic review and meta-analysis
RCTs	Randomized controlled trials
CRP	C-reactive protein
ESR	Erythrocyte sedimentation rate
TNF-alpha	Tumor necrosis factor alpha
IL	Interleukin
PGE2	Prostaglandin E2
SMD	Standard mean difference
RA	Rheumatoid arthritis
MMP-3	Matrix metalloproteinase-3
TIMP-1	Tissue inhibitor of metalloproteinase-1

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12906-025-04951-6>.

Supplementary Material 1.
Supplementary Material 2.
Supplementary Material 3.

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Clinical trial number

Not applicable.

Authors' contributions

YSH designed, and HCH conducted the study. HCH, GRH, and KHL contributed to data collection, IST contributed to statistical analysis and HCH finished the manuscript preparation. All authors reviewed the manuscript and have approved the final edited typescript.

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Data availability

Data is provided within the manuscript or supplementary information files. Further datasets used and/or analyzed in this study are available from the corresponding author upon request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Department of Medical Education, Taipei Tzu Chi Hospital, Buddhist Tzu Chi Medical Foundation, New Taipei City, Taiwan. ²Department of Physical Medicine and Rehabilitation, Taipei Tzu Chi Hospital, Buddhist Tzu Chi Medical Foundation, New Taipei City, Taiwan. ³Department of Research, Taipei Tzu Chi Hospital, Buddhist Tzu Chi Medical Foundation, New Taipei City, Taiwan. ⁴Department of Ophthalmology, Taipei Tzu Chi Hospital, Buddhist Tzu Chi Medical Foundation, New Taipei City, Taiwan. ⁵School of Medicine, Tzu Chi University, Hualien, Taiwan.

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